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BRAC UNIVERSITY
Department of Computer Science & Engineering
MID TERM EXAMINATION, Summer 2025
CSE 350: Digital Electronics and Pulse Techniques

Inspiring Excellence

Total Marks: 40 (20×2)
Time: **70 Minutes**

- There are **two(2)** questions per group. Answer any **one(1)** from **each group**.
- Figure in bracket [ ] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.

Name:	ID:	Section:

Group 1

Answer **any one** from Questions **1 & 2**

Question 1: [*CO3*]

Figure 1: A lossless digital communication

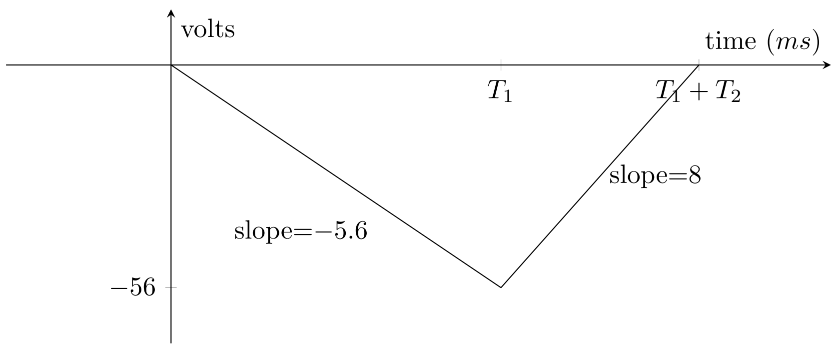

Figure 2: Characteristics of the Slow ADC

In the above communication process shown in figure 1, the **ADC** converts the V_{in} to a **4-bit** digital signal. For the characteristics graph shown in figure 2, $\boldsymbol{R = 2.5\,k\Omega}$, $\boldsymbol{C = 2\,\mu F}$.

(a)	Find the **input & reference voltage** to the **ADC**.	[5]
(b)	Find the **total conversion time** & the **binary output**.	[6]

Now, the receiver uses a **4-bit R2R DAC** with the **same** reference voltage as found in **(a)**. Also, $\boldsymbol{R_f = R = 1\,k\Omega}$.

(c)	Find the output analog voltage ($\boldsymbol{V_{out}}$) at the receiver for the digital input from the sender found in **(b)**.	[3]
(d)	Determine the **step size & maximum output** of the **DAC**.	[4]
(e)	Find the **percentage error** between V_{in} & V_{out}.	[2]

Question 2: [*CO3*]

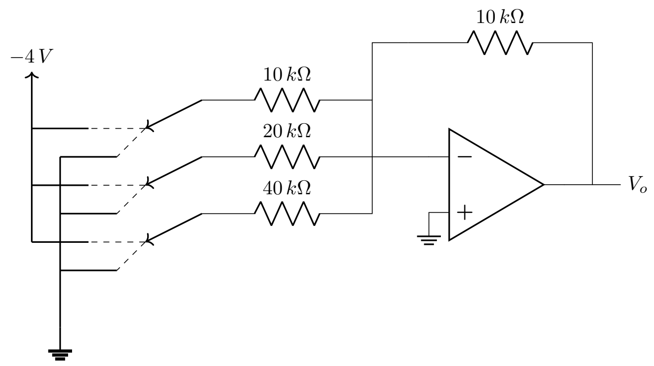

Figure 3: A 3-bit BWR DAC

The **Flash ADC** in figure 4 has a reference voltage, $\boldsymbol{V_{REF} = 16\,V}$. The output of the **DAC** in figure 3 is used as the input (V_{in}) of the ADC.

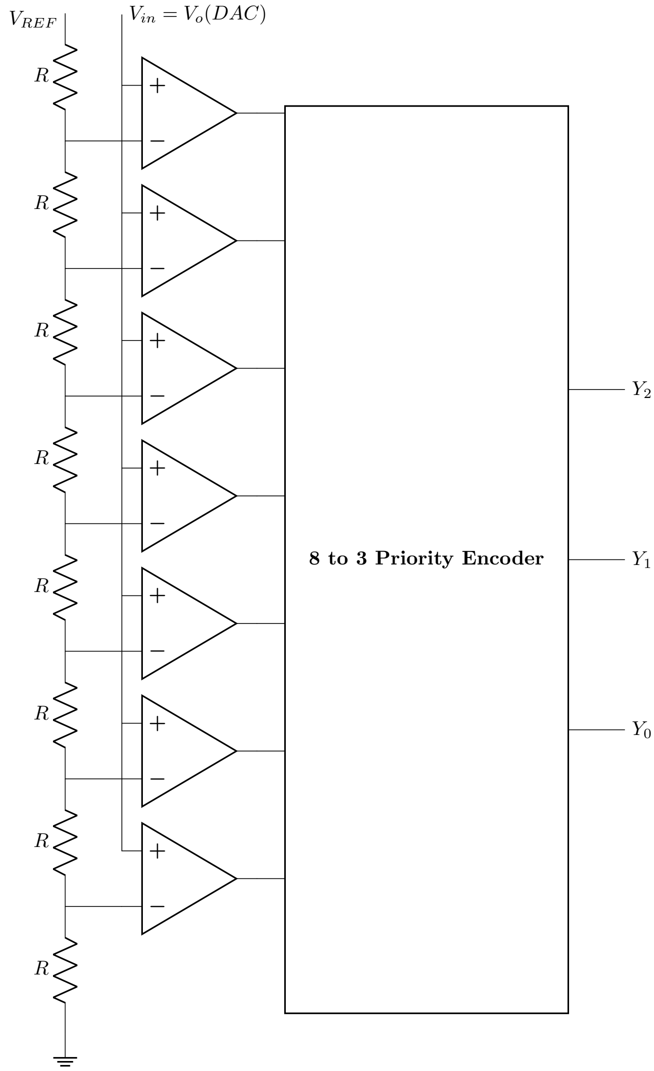

Figure 4: 3-bit Flash ADC

(a)	Determine the **resolution (step size)** of both the **BWR DAC & Flash ADC**.		[3]
(b)	(i)	Suppose the **Flash ADC** in figure 4 has an output of **'010'**. Find the **range** of $\boldsymbol{V_{in}}$ that produces this output.	[4+5]
	(ii)	Determine all possible input combinations for the DAC that will generate $\boldsymbol{V_{in}}$(V_o of DAC) in this range.	
(c)	What modifications can be made in the **DAC circuit** in figure 3, to get the **same** outputs from the ADC as the inputs to the DAC in **(b)**? [Hint: *Try making changes to equate the step sizes*]		[5]
(d)	If we wish to design a **2-bit Flash ADC** similar to the figure, comment on the hardware changes required.		[3]

Group 2

Answer **any one** from Questions **3 & 4**

Question 3: [*CO3*]

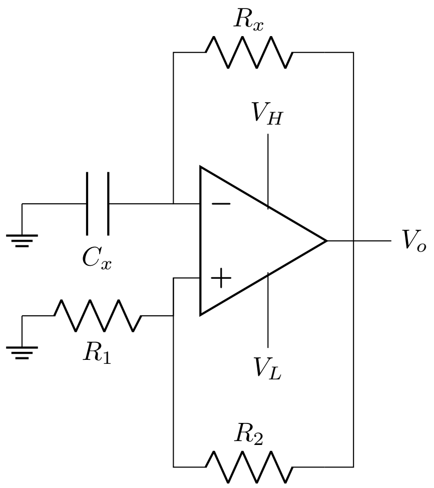

Figure 5: A Square Wave Generator

The circuit above generates a square wave with frequency **500 Hz** and **50%** duty cycle. Here, $\boldsymbol{R_1 = 2\,k\Omega}$, $\boldsymbol{R_2 = 3\,k\Omega}$, $\boldsymbol{V_H = +15\,V}$, $\boldsymbol{V_L = -15\,V}$.

(a)	Find the **time constant** & the capacitance $\boldsymbol{C_x}$ if $\boldsymbol{R_x = 22\,k\Omega}$.	[4]
(b)	Find the ratio of $\boldsymbol{R_1}$ **&** $\boldsymbol{R_2}$ to **double** the **frequency**.	[6]

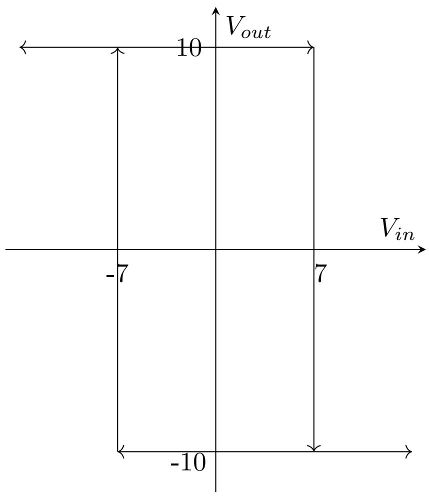

Figure 6: VTC of a Schmitt Trigger

(c)	For figure 6, find— the **type** of the Schmitt Trigger, **hysteresis width** (V_{HW}), **bias voltages**(V_H, V_L), **thresholds** (V_{TH} & V_{TL}), **shift voltage** (V_S).	[3]
(d)	Design a Schmitt Trigger with the same VTC, but **shifted by +3 V**. Draw the new **VTC** and the **circuit**. [Hint: *Find the ratio of* R_1 &R_2 *first*]	[5]
(e)	Apply the input $\boldsymbol{12\,sin(2\pi ft)}$ to the Schmitt Trigger & VTC designed in **(d)** and plot the V_{out}.	[2]

Question 4: [*CO3*]

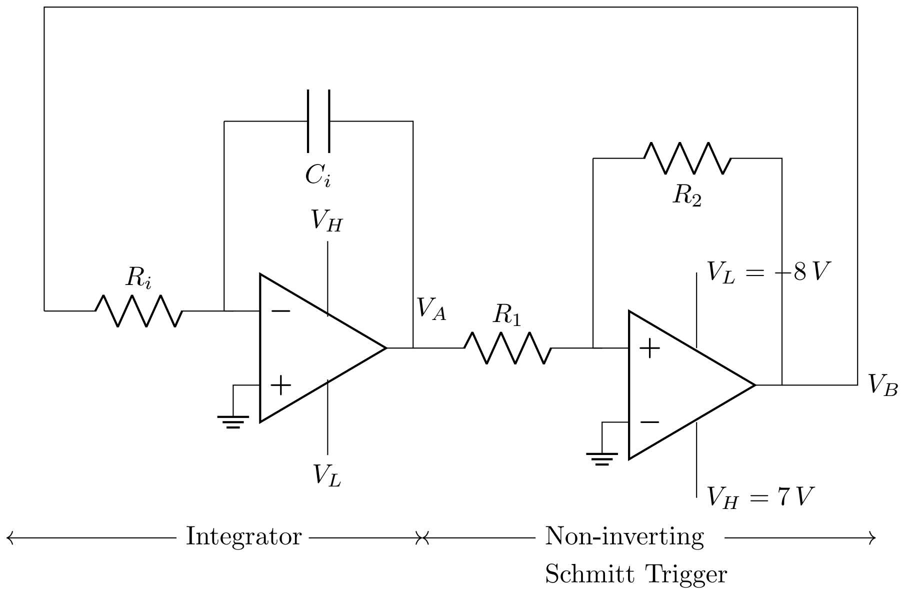

Figure 7: A Triangular Wave Generator

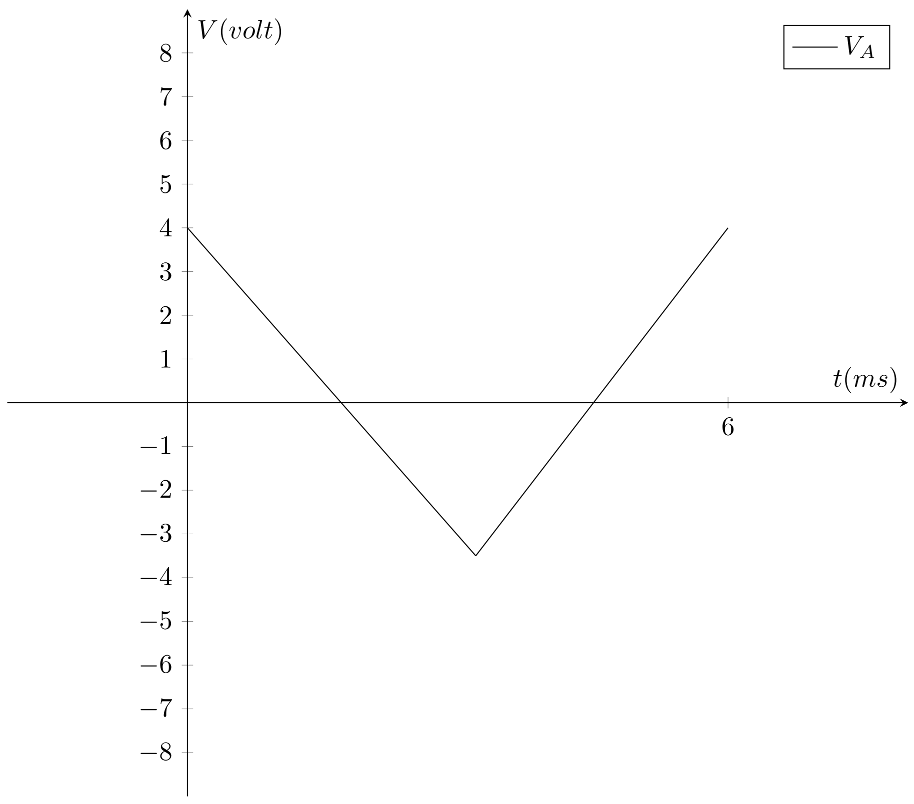

Figure 8: Output of the Triangular Wave Generator

For the triangular wave shown—

(a)	Find the **time period & duty cycle**. If $R_i = 2\,k\Omega$, find $\boldsymbol{C_i}$.	[8]
(b)	Draw the corresponding **square wave**, output from the non-inverting Schmitt Trigger of the triangular wave generator (*on figure 8*). Find its **duty cycle**.	[5]
(c)	Draw the **VTC** of the Schmitt Trigger in figure 7.	[2]
(d)	Consider a separate Schmitt Trigger has the similar VTC as in **(c)**, but shifted by $\boldsymbol{+2\,V}$. **Design** the circuit and draw the **output vs time** graph if the input $\boldsymbol{V_i = 5sin(\omega t)}$ was applied to it.	[5]